A Provably Secure, Efficient, and Flexible Authentication Scheme for Ad hoc Wireless Sensor Networks

Chin-Chen Chang, *Fellow, IEEE*, and Hai-Duong Le

***Abstract*—In 2014, Turkanovic *et al.* proposed a smart card-based authentication scheme for heterogeneous ad hoc wireless sensor network. This scheme is very efficient since it employs only hash function and XOR operation. However, we found that Turkanovic *et al.*'s scheme is vulnerable to impersonation attack with node capture, stolen smart card attack, sensor node spoofing attack, stolen verifier attack, and fails to ensure backward secrecy. We propose an efficient scheme to overcome all those weaknesses. Moreover, we also propose an advanced scheme, which provides perfect forward secrecy without much modification from the first proposed scheme.**

***Index Terms*—Authentication, elliptic curve cryptography, key agreement, wireless sensor networks.**

I. Introduction

WIRELESS Sensor Networks (WSNs) are cost-effective solutions for a wide range of real-time monitoring applications, such as traffic monitoring, environmental monitoring, wildlife monitoring, homeland security, health care, etc. They are normally deployed in unattended environments, which are sometimes under hostile conditions. Besides monitoring, they are also used for controlling equipment in manufactures, battlefield weapons, etc.

A Wireless Sensor Network is comprised of a large number of specialized and autonomous sensors communicating over wireless network. A sensor node is typically constrained by its low memory, low battery power, low bandwidth and limited computational ability. Therefore, it is desirable not to exert sensor nodes with heavy workloads.

In WSNs, data collected by sensor nodes sometimes contain valuable and confidential information that only authorized users are allowed to access. Moreover, in the case where a user commands a sensor node to perform certain tasks, the user must be authenticated before sending instructions to the sensor node. There are two approaches in authenticating users in WSNs: (a) a user is authenticated by the gateway node (GWN) before accessing sensor nodes; (b) a user directly contacts a sensor node and performs authentication with it.

Since sensor nodes are limited in terms of computation and communication capabilities, lightweight authentication and key agreement protocols [1]–[11] are preferred for WSNs. In 2006, Wong et al. [2] proposed a dynamic strong-password based user authentication for WSN. This protocol is considered to be lightweight since it uses only simple operations such as hash functions and exclusive-OR operations. However, in 2007, Tseng et al. [3] showed that Wong et al.'s protocol is vulnerable to replay and forgery attacks, and any sensor node could reveal the users' passwords. Furthermore, Das [4] found that Wong et al.'s protocol also suffers the same login-id threat and the stolen-verifier attack. Das then presented a two-factor authentication scheme using smart card in which users are authenticated by gateway nodes. Later studies [5]–[7] revealed that Das's scheme fails to provide mutual authentication and key agreement, and it is vulnerable to many attacks (e.g. insider attack, impersonation attack, node-capture attack, denial-of-service attack, etc.). In 2010, Khan et al. [5] proposed a protocol as an improvement of Das's, but Vaidya et al. [8] showed that it is susceptible to stolen smart card attack and impersonation attack.

In 2012, two lightweight smart card-based authentication protocols were proposed by Das et al. [9] and Xue et al. [10] separately. However, Turkanovic and Holbl [12] demonstrated that Das et al.'s scheme has flaws that make it infeasible for real-life implementation. In 2013, Li et al. [11] showed that Xue et al.'s protocol suffers several attacks (e.g. insider attack, stolen verifier attack, many logged-in users attack, etc.) and proposed an advanced scheme that eliminates those vulnerabilities in Xue et al.'s.

Recently, in 2014, Turkanovic et al. [1] proposed a new smart card based authentication scheme for heterogeneous ad hoc wireless sensor networks in which a user can contact and authenticate directly with a sensor node. Although it is an efficient scheme which employs only hash function and exclusive-OR operation, we found that Turkanovic et al.'s protocol is susceptible to stolen smart card attack, impersonation attack with node capture, sensor node spoofing attack, stolen verifier attack, and it fails to ensure backward secrecy.

In those lightweight protocols, a sensor node is always provided with a secret value which is either computable only by the gateway or pre-shared with the gateway node. If an adversary eavesdrops the communication channel and obtains this secret

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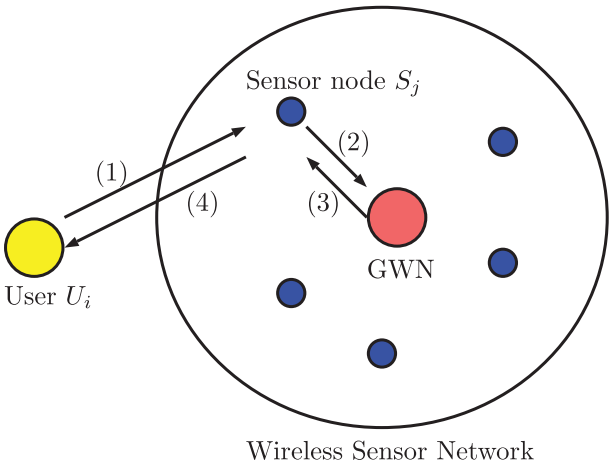


Fig. 1. Communication model.

value from a sensor node, it can compute the previous session keys of that sensor node. Since sensor nodes are often deployed in hostile environment, there is high possibility that an adversary captures a sensor node and obtains the aforementioned secret value. Thus, the lightweight protocols cannot provide perfect forward secrecy under node capture attack. A remedy for this problem is to employ public key cryptosystems, such as Elliptic Curve Cryptography (ECC), which demand more resources in terms of computation cost.

In 2011, Yeh et al. [13] proposed a first authentication protocol using both ECC and smart card for WSNs. However, in 2013, Shi and Gong [15] pointed out that Yue et al.'s protocol fails to provide perfect forward secrecy and mutual authenticated key agreement between the user and the sensor node. They then proposed an ECC-based authentication protocol using smart card to overcome those weaknesses. But Choi et al. [16] found that Shi and Gong's scheme is vulnerable to session key attack, stolen smart card attack, and sensor exhausting attack.

In this paper, we first demonstrate the weaknesses in Turkanovic et al.'s protocol [1]. Then, we propose a flexible authentication protocol using smart card for WSNs which operates in two modes. The first mode provides a lightweight authentication scheme which overcomes the weaknesses in Turkanovic et al.'s. The second mode is an advanced protocol based on ECC which provides perfect forward secrecy. Depending on application and the security level required, the user can choose which mode to be used. The differences between the two modes are small regarding to implementation and storage cost.

II. Preliminaries

A. Communication Model

Suppose that a user U_i wants to establish a secure communication with a sensor node S_j in a Wireless Sensor Network. The communication model of the authentication and key establishment protocol is shown in Figure 1. U_i first sends a login message (1) directly S_j which subsequently requests (2) the gateway node GWN for authenticating the user. When receiving positive response (3) from GWN, the sensor node accepts U_i and replies with key establishment information (4).

B. Security Model

In this section, we describe our security model following Abdalla et al.'s Real-Or-Random (ROR) model [17].

Participants. Let the instance t of the sensor node S_j be $\Pi^t_{S_j}$, the instance u of the user ID_i be $\Pi^u_{U_i}$, and the instance v of the gateway node GWN be Π^v_{GWN}; these instances are called oracles.

Partnering. Partnering is defined by session ID and partner ID of a user U_i or a sensor node S_j [18]. The partner of an instance $\Pi^u_{U_i}$ of the user U_i is the instance $\Pi^t_{S_j}$ of a sensor node S_j, and vice versa. The instance $\Pi^t_{S_j}$ is called the partner ID $pid^u_{U_i}$ of $\Pi^u_{U_i}$. The partial transcript of all the messages exchanged between the user U_i and the sensor node S_j is unique and called a session ID ($sid^u_{U_i}$) for the current session in which the instance $\Pi^u_{U_i}$ participates.

Freshness. The instance $\Pi^u_{U_i}$ or $\Pi^t_{S_j}$ is *fresh* if their session key SK has not revealed to the adversary $\mathcal{A}$.

Adversary. The adversary $\mathcal{A}$ is assumed to have complete control over all the communications in a WSN. Besides the ability to read and modify all the exchanged messages, $\mathcal{A}$ can also fabricate new messages and inject them into the network. The adversary is provided the following queries:

- $Execute(\Pi^t, \Pi^u)$: The adversary executes this query to obtain the messages transmitted between two honest participants; this query models an eavesdropping attack.
- $Send(\Pi^t, m)$: This query models an active attack in which $\mathcal{A}$ sends a message m to a participant instance Π^t and receives a response message.
- $CorruptSC(\Pi^l_{U_i})$: This query models the smart card lost attack. It outputs the information stored in the smart card.
- $CorruptSensor(\Pi^t_{S_j})$: This query models the attack in which the long-term secret key of the sensor node S_j is compromised. This query is used to illustrate the perfect forward secrecy property of the protocol.
  $CorruptSC$ and $CorruptSensor$ queries follow the weak-corruption model in which the ephemeral keys and internal data of the participant instances are not corrupted.
- $Test(\Pi^t)$: This query models the semantic security of the session key following the Indistinguishability in the Real-Or-Random (ROR) model [17]. A coin b is flipped at the start of the experiment, its value is kept secret from the adversary $\mathcal{A}$ and used to decide the output of $Test()$ query. When the adversary executes this query, if the session key has been established and is *fresh*, the instance Π^t returns the session key if $b = 1$ or a random number in the same domain if $b = 0$; otherwise, it returns $\perp$.

Semantic security of the session key. In the Real-Or-Random model, the adversary is challenged in an experiment to distinguish between an instance's real session key and a random key. The adversary can ask many $Test$ queries to either the user instance or the sensor node instance. The output of $Test$ must be consistent with regard to the random bit b. In the end, $\mathcal{A}$ returns a guess bit b' and wins the game if $b' = b$.

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Let *Succ* denote the event in which the adversary wins the game. The advantage of $\mathcal{A}$ in breaking the semantic security of the protocol $\mathcal{P}$ is $Adv_{\mathcal{P}}^{ake} = |2 \cdot Pr[Succ] - 1|$. We say that the protocol $\mathcal{P}$ is a secure authentication and key agreement protocol in the ROR sense if $Adv_{\mathcal{P}}^{ake}(\mathcal{A})$ is negligible.

Random Oracle. In this paper, the participants and the adversary are provided with a hash function $h()$ which is modelled by a random oracle. Let $Hash$ denote the random oracle and it is simulated by a two-tuple (x, k) table of binary strings. When one makes a hash query $h(x)$, the random oracle returns k if x is found; otherwise, it returns a uniformly random string k and stores the pair (x, k) into the table.

C. Elliptic Curve Discrete Logarithm Problem

Let $q > 3$ be a large prime and $E_{a,b}$ denote the group of points of the elliptic curve $E_{a,b} : y^2 = x^3 + ax + b$ over the finite field $\mathbb{F}_q$, where $a, b \in \mathbb{F}_q$ satisfy $4a^3 + 27b^2 \neq 0 \bmod q$. Let $G_p = \langle P \rangle$ be a cyclic group of prime order p, where P is a generator point of the group and $P \in E_{a,b}(\mathbb{F}_q)$. Then, the three mathematical problems [20] in Elliptic Curve Cryptography (ECC) are defined as follows:

- The *discrete logarithm* (ECDL) problem is to find a when given a point $Q \in G_p$, where $a \in \mathbb{Z}_p$ and $Q = aP$.
- The *computational Diffie-Hellman* (ECCDH) problem is to find abP when given aP and bP, where $a, b \in \mathbb{Z}_p$.
- The *decisional Diffie-Hellman* (ECDDH) problem is to decide whether $abP = cP$ when given aP, bP and cP, where $a, b, c \in \mathbb{Z}_p$.

It is clearly that ECCDH is not harder than ECDL, and ECDDH is not harder than ECCDH. If ECDDH problem is intractable, then there is no polynomial time adversary can solve ECDDH, ECCDH, and ECDL problems with non-negligible probability.

III. Related Work

A. Review of Turkanovic et al.'s Scheme

In this section, we review Turkanovic et al.'s authentication and key agreement scheme for heterogeneous ad hoc wireless sensor networks in brief. The scheme has totally five main phases: pre-deployment phase, registration phase, login phase, authentication phase, and password-change phase. The notations used in this paper are listed in Table I.

1) Pre-deployment Phase: Suppose there are m regular sensor nodes deployed in a wireless sensor network. Each of the sensor node S_j, where $1 \leq j \leq m$, is assigned an identity SID_j and a randomly-generated secret X_{GWN-S_j}. The gateway node GWN stores SID_j and X_{GWN-S_j} in its memory.

2) Registration Phase: In this phase, users and sensor nodes register with the gateway node GWN in different procedures as shown in Figure 2 and 3, separately. The registration procedure for a user U_i consists of the following steps:

Step 1 The user U_i selects an identity ID_i and a password PW_i, and chooses a random number r_i. U_i computes $MP_i = h(r_i \| PW_i)$, $MI_i = h(r_i \| ID_i)$, and sends the message $m_{Ureg} = \{MI_i, MP_i\}$ to GWN via a secure channel.

TABLE I
Notations

U_i	The user
ID_i	The identity of U_i
PW_i	The password of U_i
SC_i	The smart card of U_i
GWN	The gateway node
S_j	The sensor node
SID_j	The identity of S_j
X_{GWN}	The long-term secret of GWN
X_{GWN-U_i}	The secret shared between GWN and U_i
X_{GWN-S_j}	The secret shared between GWN and S_j
T_1, T_2, T_3, T_4	The timestamps
ΔT	The expected transmission delay
r_i, r_i', r_j, K_i, K_j, a, b	The random numbers
P	A point on the elliptic curve
$P.x$	The x-axis value of the point P
$\|\|$, $\oplus$, $h()$	The concatenation, XOR, and hash operation

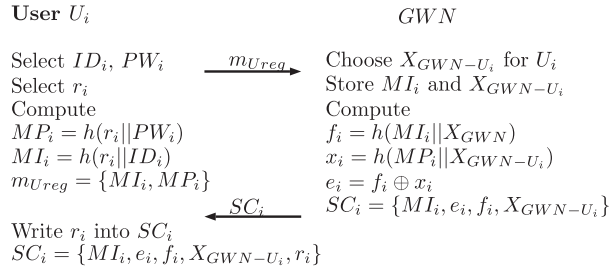


Fig. 2. Registration phase for a user U_i.

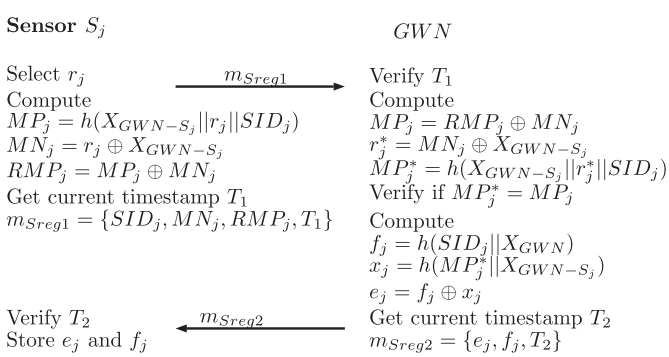


Fig. 3. Registration phase for a sensor node S_j.

Step 2 GWN chooses a secret X_{GWN-U_i} for U_i, then stores MI_i and X_{GWN-U_i} into its memory. It computes $f_i = h(MI_i \| X_{GWN})$, $x_i = h(MP_i \| X_{GWN-U_i})$, $e_i = f_i \oplus x_i$, and issues the smart card SC_i containing $\{MI_i, e_i, f_i, X_{GWN-U_i}\}$ to U_i.

Step 3 U_i writes r_i into SC_i.

The detailed steps of the registration procedure for a sensor node S_j are described as follows.

Step 1 S_j selects a random nonce r_j, and computes $MP_j = h(X_{GWN-S_j} \| r_j \| SID_j)$, $MN_j = r_j \oplus X_{GWN-S_j}$, $RMP_j = MP_j \oplus MN_j$, then sends $m_{Sreg1} = \{SID_j, MN_j, RMP_j, T_1\}$ to GWN via an unsecure channel.

Step 2 If T_1 is valid, GWN computes $MP_j = RMP_j \oplus MN_j$, $r_j^* = MN_j \oplus X_{GWN-S_j}$, and $MP_j^* = h(X_{GWN-S_j} \| r_j \| SID_j)$. If $MP_j^* = MP_j$, GWN computes $f_j = h(SID_j \| X_{GWN})$, $x_j =$

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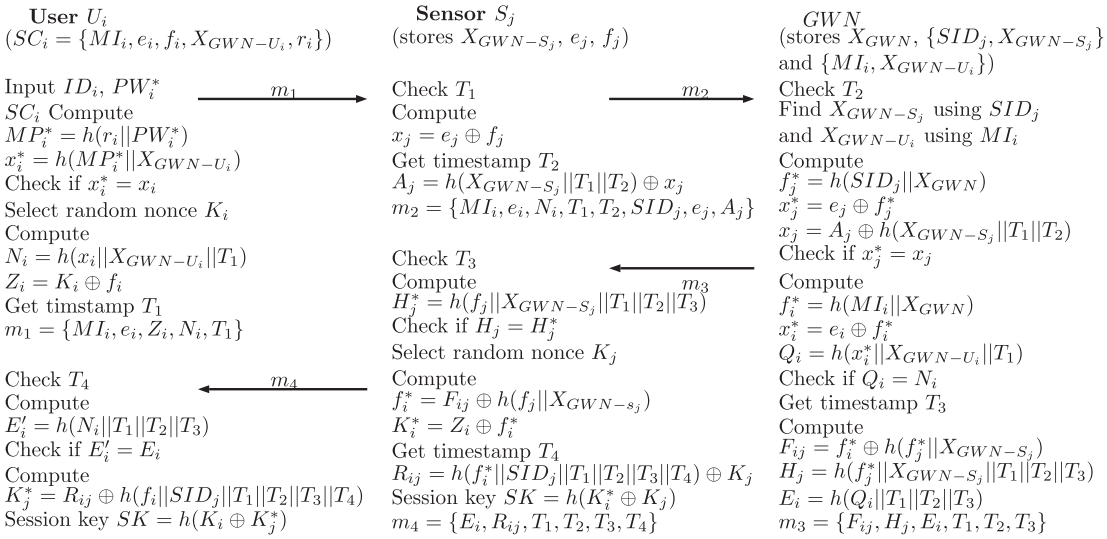


Fig. 4. Login and authentication phase.

$h(MP_j^* \| X_{GWN-S_j})$, $e_j = f_j \oplus x_j$, and sends $m_{Sreg2} = \{e_j, f_j, T_2\}$ to S_j.

Step 3 If T_2 is valid, S_j has succeeded in registering to GWN. Finally, it stores e_j and f_j into its memory.

3) Login Phase: In this phase, U_i selects the sensor node S_j that it wants to communicate with and performs the login procedure as follows. U_i inputs the password PW_i^*. SC_i computes $MP_i^* = h(r_i \| PW_i^*)$, $x_i^* = h(MP_i^* \| X_{GWN-U_i})$. If $x_i^* = x_i$, SC_i selects a random nonce K_i, and computes $N_i = h(x_i \| X_{GWN-U_i} \| T_1)$, $Z_i = K_i \oplus f_i$. Then, it sends the login request $m_1 = \{MI_i, e_i, Z_i, N_i, T_1\}$ to S_j.

4) Authentication Phase: After receiving the login message m_1 from U_i, the sensor node S_j authenticates U_i in the following procedure as illustrated in Figure 4.

Step 1 If T_1 is valid, S_j computes $x_j = e_j \oplus f_j$, $A_j = h(X_{GWN-S_j} \| T_1 \| T_2) \oplus x_j$, then sends $m_2 = \{MI_i, e_i, N_i, T_1, T_2, SID_j, e_j, A_j\}$ to GWN.

Step 2 GWN checks T_1 and finds the secret X_{GWN-S_j} corresponding to SID_j in its memory. It computes $f_j^* = h(SID_j \| X_{GWN})$, $x_j^* = e_j \oplus f_j^*$, $x_j = A_j \oplus h(X_{GWN-S_j} \| T_1 \| T_2)$. If $x_j^* = x_j$, GWN has successfully authenticated S_j; otherwise, it terminates the session.

After that, GWN computes $f_i^* = h(MI_i \| X_{GWN})$, $x_i^* = e_i \oplus f_i^*$, and $Q_i = h(x_i^* \| X_{GWN-U_i} \| T_1)$, where X_{GWN-U_i} for the corresponding MI_i is retrieved from its memory. If $Q_i = N_i$, U_i is authentic. GWN computes $H_j = h(f_j^* \| X_{GWN-S_j} \| T_1 \| T_2 \| T_3)$, $F_{ij} = f_i^* \oplus h(f_j^* \| X_{GWN-S_j})$, $E_i = h(Q_i \| T_1 \| T_2 \| T_3)$, and sends $m_3 = \{F_{ij}, H_j, E_i, T_1, T_2, T_3\}$ to S_j.

Step 3 If T_3 is valid and $H_j = h(f_j \| X_{GWN-S_j} \| T_1 \| T_2 \| T_3)$, S_j has successfully authenticated both GWN and U_i, and it selects a random nonce K_j. Then, it computes $f_i^* = F_{ij} \oplus h(f_j \| X_{GWN-S_j})$, $K_i^* = Z_i \oplus f_i^*$, $R_{ij} = h(f_i^* \| SID_j \| T_1 \| T_2 \| T_3 \| T_4) \oplus K_j$, the session key $SK = h(K_i \oplus K_j)$, and sends the message $m_4 = \{E_i, R_{ij}, T_1, T_2, T_3, T_4\}$ to U_i.

Step 4 If T_4 is valid and $E_i = h(N_i \| T_1 \| T_2 \| T_3)$, U_i trusts that S_j is genuine. SC_i computes $K_j = R_{ij} \oplus h(f_i \| SID_j \| T_1 \| T_2 \| T_3 \| T_4)$ and the session key $SK = h(K_i \oplus K_j)$.

5) Password-change Phase: U_i inputs the current password PW_i^{OLD}. SC_i computes $MI_i^* = h(r_i \| PW_i^{OLD})$ and $x_i^* = f_i \oplus e_i$. If $x_i^* = x_i$, U_i inputs the new password PW_i^{NEW}. SC_i computes $MP_i^{**} = h(r_i \| PW_i^{NEW})$, $x_i^{NEW} = h(MP_i^{**} \| X_{GWN-U_i})$, and $e_i^{NEW} = f_i \oplus x_i^{NEW}$. Finally, SC_i replaces the old value e_i with the new value e_i^{NEW} in its memory.

B. Weaknesses of Turkanovic et al.'s

1) Stolen Smart Card and Smart Card Breach Attack: In this attack, an adversary $\mathcal{A}$ first obtains a smart card of a legitimate user, and it uses the information stored in the smart card's memory to gain access to any sensor node. Suppose $\mathcal{A}$ can get the smart card SC_i of the user U_i. The adversary can log into the sensor node S_j by performing the following steps.

Step 1 First, $\mathcal{A}$ extracts the values MI_i, e_i, f_i, and X_{GWN-U_i} from SC_i's memory.

Step 2 Next, it selects a random nonce K_i and computes $x_i = e_i \oplus f_i$, $N_i = h(x_i \| X_{GWN-U_i} \| T_1)$, and $Z_i = K_i \oplus f_i$.

Step 3 It then constructs $m_1 = \{MI_i, e_i, N_i, T_1\}$ and sends it to the sensor node S_j. The message m_1 is valid and will be accepted by GWN and S_j.

Step 4 When receiving $m_4 = \{E_i, R_{ij}, T_1, T_2, T_3, T_4\}$ from S_j, $\mathcal{A}$ computes $K_j = R_{ij} \oplus h(f_i \| SID_j \| T_1 \| T_2 \| T_3 \| T_4)$ and the session key $SK = K_i \oplus K_j$. In the end, the adversary has successfully logged into S_j.

2) Impersonation Attack with Sensor Node Capture: Turkanovic et al.'s scheme is vulnerable to the impersonation attack in which an adversary $\mathcal{A}$ masquerades as a legitimate user and gains access to one or many sensor nodes simultaneously after compromising one of them. Suppose $\mathcal{A}$ has compromised the sensor node S_j and used S_j to authenticate the user U_i once.

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After that, every time U_i sends a login message m_1 to a sensor node, $\mathcal{A}$ captures this message and uses it to log into any sensor node of its choice. The detailed description of this attack is shown below.

Step 1 The adversary first compromises the sensor node S_j.

Step 2 When U_i sends login request $m_1 = \{MI_i, e_i, Z_i, N_i, T_1\}$ to the compromised sensor node S_j for the first time, $\mathcal{A}$ controls S_j to authenticate U_i normally. In Step 3 of the authentication phase, S_j computes $f_i = F_{ij} \oplus h(f_j \| X_{GWN-S_j})$; thus, $\mathcal{A}$ can acquire the values e_i and f_i from S_j.

Step 3 Suppose the adversary wants to gain access to a sensor node S_t whose identity is SID_t. It first monitors the communication channel and looks for a new login request from U_i. When U_i sends $m_1 = \{MI_i, e_i, Z_i, N_i, T_1\}$ to any sensor node again, $\mathcal{A}$ immediately intercepts and forwards m_1 to S_t.

Step 4 Upon receiving m_1 forwarded by $\mathcal{A}$, the sensor node S_t contacts the gateway node GWN to authenticate U_i as described in Steps 1 and 2 of the authentication phase. Since all the parameters MI_i, e_i, $Z_i = K_i \oplus f_i$ and $N_i = h(x_i \| X_{GWN-U_i} \| T_1)$, and T_1 in m_1 are not bound to SID_t, the sensor node S_t will be deceived that m_1 was intentionally sent to it, and authenticate U_i as usual. After both GWN and S_t have successfully authenticated U_i, S_t chooses a random nonce K_t and computes $R_{it} = h(f_i \| SID_t \| T_1 \| T_2 \| T_3 \| T_4) \oplus K_t$ and the session key $SK = K_i \oplus K_t$. Then, it sends the reply message $m_4 = \{E_i, R_{ij}, T_1, T_2, T_3, T_4\}$ to U_i. At this point, the adversary intercepts and seizes the message m_4. $\mathcal{A}$ then uses the value f_i obtained in Step 2 above to compute $K_i = Z_i \oplus f_i$, $K_t = R_{it} \oplus h(f_i \| SID_t \| T_1 \| T_2 \| T_3 \| T_4)$, and $SK = h(K_i \oplus K_t)$.

Finally, the adversary $\mathcal{A}$ has successfully impersonated U_i to log into S_t and share the session key SK with it. In addition, $\mathcal{A}$ can perform Steps 3 and 4 above on more than one sensor node simultaneously and gain access to all of them.

3) Sensor Node Spoofing Attack with Sensor Node Capture: After compromising one sensor node, an adversary $\mathcal{A}$ can masquerade as any legitimate sensor node to which a user is trying to log in. Suppose $\mathcal{A}$ has already compromised the sensor node S_j. When the user U_i wants to log into the sensor node S_t, the adversary performs the spoofing attack in the following steps:

Step 1 After compromising the sensor node S_j, the adversary acquires the parameters X_{GWN-S_j}, SID_j, e_j, and f_j from S_j's memory.

Step 2 When U_i sends $m_1 = \{MI_i, e_i, Z_i, N_i, T_1\}$ to S_t, the adversary intercepts that message. Next, it computes $x_j = e_j \oplus f_j$ and $A_j = h(X_{GWN-S_j} \| T_1 \| T_2) \oplus x_i$, and sends $m_2 = \{MI_i, e_i, N_i, T_1, T_2, SID_j, e_j, A_j\}$ to the gateway node.

Step 3 Upon receipt of m_2 from $\mathcal{A}$, the gateway node GWN performs all the checking and computing as in Step 2 of the authentication phase. Since MI_i, e_i, and N_i are not bound to S_k, the gateway does not know that they are sent to S_k, not to S_j. Furthermore, $\mathcal{A}$ has used the valid parameters X_{GWN-S_j}, e_i, and f_i to compute A_j. Thus, GWN trusts that m_2 is valid and computes F_{ij}, E_i and H_j before sending the reply message $m_3 = \{F_{ij}, H_j, E_i, T_1, T_2, T_3\}$ to $\mathcal{A}$, who is masquerading as S_j now.

Step 4 After receiving m_3 from GWN, the adversary computes $f_i = F_{ij} \oplus h(f_j \| X_{GWN-S_j})$ and $K_i = Z_i \oplus f_i$. It then chooses a random nonce K_t and computes $R_{ij} = h(f_i \| SID_t \| T_1 \| T_2 \| T_3 \| T_4)$ and $SK = K_i \oplus K_t$. Finally, it sends $m_4 = \{E_i, R_{ij}, T_1, T_2, T_3, T_4\}$ to U_i.

Step 5 U_i first verifies E_i, then it believes that it is communicating with a legitimate sensor node. Next, it computes $K_t = R_{ij} \oplus h(f_i \| SID_t \| T_1 \| T_2 \| T_3 \| T_4)$ and the session key $SK = K_i \oplus K_t$.

In the end, $\mathcal{A}$ has succeeded in masquerading as the sensor node S_t.

4) Fail to Achieve Forward and Backward Secrecy: In Turkanovic et al., the session key is computed as $SK = K_i \oplus K_j$, where $K_i = Z_i \oplus f_i$ and $K_j = R_{ij} \oplus h(f_i \| SID_j \| T_1 \| T_2 \| T_3 \| T_4)$. Therefore, if an adversary $\mathcal{A}$ could obtain f_i, it can compute the session key SK between U_i and any sensor node since Z_i and R_{ij} are transmitted over unsecure channel. In the three attacks above (smart card stolen attack, impersonation attack and sensor spoofing attack), the adversary can always acquire f_i. Therefore, Turkanovic et al.'s scheme fails to provide both forward and backward secrecy.

5) Stolen Verifier Attack: The straightforward way for an adversary to masquerade as a sensor node S_j is obtaining e_j, f_j and X_{GWN-S_j}. The two values e_j and f_j are delivered to S_j from GWN in the registration phase over an unsecure network; thus, $\mathcal{A}$ can eavesdrop and capture them. The secret X_{GWN-S_j} is stored in the memory of the gateway node. Therefore, if the adversary can break in GWN's memory and retrieve X_{GWN-S_j}, it will be able to masquerade S_j.

IV. The Proposed Schemes

In this section, we propose a new protocol $\mathcal{P}_1$ which overcomes the weaknesses in Turkanovic et al.'s scheme. Moreover, the proposed scheme requires no database at the gateway node to keep the shared secrets. The proposed scheme has four phases: pre-deployment phase, registration phase, authentication phase, and password changing phase.

Since $\mathcal{P}_1$ is lightweight, it does not provide perfect forward secrecy. Therefore, we also present a protocol $\mathcal{P}_2$ that can guarantee perfect forward secrecy. $\mathcal{P}_2$ is an advanced version of $\mathcal{P}_1$ and shares with $\mathcal{P}_1$ the same procedures in pre-deployment phase, registration phase, and password changing phase. We can think of $\mathcal{P}_1$ and $\mathcal{P}_2$ as two operation modes with different security levels in an authentication scheme.

A. The Protocol $\mathcal{P}_1$

In this phase, the gateway node GWN is provided a long-term secret X_{GWN}, which is generated randomly. Unlike Turkanovic et al.'s scheme, GWN does not require storing any shared secrets for the sensor nodes and users.

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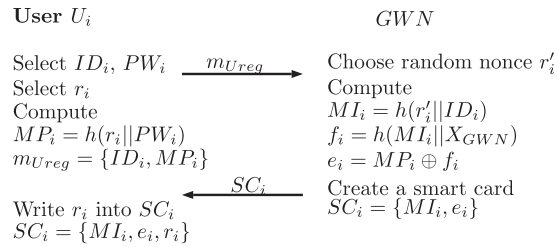


Fig. 5. Registration phase.

As for a sensor node S_j, it is first assigned an identity SID_j. Then, the gateway computes $f_j = h(SID_j \| X_{GWN})$, and the parameters SID_j and f_j are written into the sensor node's memory before deploying S_j into the network.

1) Registration Phase: Before gaining access to the sensor nodes of a wireless sensor network, a user U_i needs to register with the gateway node as shown in Figure 5.

Step 1 U_i selects an identity ID_i and a password PW_i, and generates a random number r_i. The user then computes $MP_i = h(r_i \| PW_i)$ before sending $m_{Ureg} = \{ID_i, MP_i\}$ to the gateway node GWN.

Step 2 Upon receipt of m_{Ureg}, the gateway node verifies the user's identity and generates a random number r_i'. Next, GWN computes the pseudo-identity $MI_i = h(r_i' \| ID_i)$ for U_i, and uses its long-term secret X_{GWN} to compute $f_i = h(MI_i \| X_{GWN})$, and $e_i = MP_i \oplus f_i$. Lastly, GWN personalises a smart card SC_i containing $\{MI_i, e_i\}$ and issues it to U_i.

Step 3 After receiving the smart card SC_i, the user U_i writes the value r_i into it. Eventually, the smart card SC_i contains $\{MI_i, e_i, r_i\}$.

2) Authentication Phase: When a user U_i wants to log into a sensor node S_j, U_i performs the following steps as shown in Figure 6.

Step 1 The user U_i inserts the smart card SC_i into a terminal, and inputs the identity ID_i and the password PW_i. The smart card computes $MP_i = h(r_i \| PW_i)$, $f_i = e_i \oplus MP_i$ and $Y_i = h(f_i \| T_1)$, whereby T_1 represents the current timestamp of SC_i. Next, SC_i chooses a random nonce K_i and computes $Z_i = K_i \oplus Y_i$, and $N_i = h(Y_i \| MI_i \| SID_j)$. Afterwards, the smart card sends the login request $m_1 = \{MI_i, Z_i, N_i, T_1\}$ to the sensor node S_j.

Step 2 When receiving m_1 from U_i, the sensor node S_j verifies the timestamp T_1 by checking if $|T_1 - T_c| \leq \Delta T$, where T_c is the current timestamp at S_j and ΔT is the allowed transmission delay. If T_1 is valid, S_j gets the current timestamp T_2 and computes $A_j = h(f_j \| N_i \| T_2)$, where f_j is in the node's memory; otherwise, S_j rejects the login request from U_i. After that, the sensor node sends the message $m_2 = \{MI_i, N_i, SID_j, A_j, T_1, T_2\}$ to the gateway node GWN.

Step 3 Upon receipt of m_2 from S_j, the gateway node verifies both T_1 and T_2. If either T_1 or T_2 is invalid, GWN terminates the session. If both of them are valid, the gateway uses its long-term secret X_{GWN} to compute $f_j' = h(SID_j \| X_{GWN})$, $f_i' = h(MI_i \| X_{GWN})$, $A_j' = (f_j' \| N_i \| T_2)$, $Y_i' = h(f_i' \| T_1)$, and $N_i' = h(Y_i' \| MI_i \| SID_j)$. Next, GWN verifies A_j and N_i in m_2. If $N_i' = N_i$ and $A_j' = A_j$, the gateway accepts U_i; otherwise, it rejects U_i. After successfully authenticating U_i, GWN computes $F_{ij} = Y_i' \oplus h(f_j' \| T_3)$, $H_j = h(Y_i')$, and $E_i = h(f_i' \| N_i')$ and sends $m_3 = \{F_{ij}, H_j, E_i, T_3\}$ to S_j.

Step 4 Once receiving m_3, the sensor node S_j verifies T_3. If $|T_3 - T_c| \leq \Delta T$, S_j computes $Y_i' = F_{ij} \oplus h(f_j \| T_3)$ and $H_j' = h(Y_i')$. If $H_j' = H_j$, the sensor node accepts U_i; otherwise, it rejects U_i. Then, it selects a random nonce K_j and computes $K_i' = Z_i \oplus Y_i'$, $R_{ij} = h(K_i' \| T_4) \oplus K_j$, whereby T_4 is the current timestamp. After sending the message $m_4 = \{R_{ij}, E_i, T_4\}$ to U_i, the sensor node computes the session key $SK = h(K_i' \oplus K_j)$.

Step 5 After receiving the reply message m_4 from S_j, the smart card SC_i tests T_4. If $|T_4 - T_c| \leq \Delta T$, it computes $E_i' = h(f_i \| N_i)$ and checks whether $E_i' = E_i$. If they are equal, SC_i believes that S_j is legitimate and computes $K_j' = R_{ij} \oplus h(K_i \| T_4)$. Then, SC_i computes the session key $SK = h(K_i \oplus K_j')$. The validity of the session key SK is implicitly verified. If SC_i and S_j exchange data successfully over the secure channel constructed using SK, the computed session key is valid; otherwise, U_i terminates the session.

3) Password Changing Phase: Assuming the user U_i is already authenticated. In order to change password, U_i first inserts the smart card into a terminal and inputs the identity ID_i and the password PW_i. The smart card computes $MP_i = h(r_i \| PW_i)$ and requests U_i to input a new password PW_i^{new}. After that, SC_i computes $MP_i^{new} = h(r_i \| PW_i^{new})$ and $e_i^{new} = e_i \oplus MP_i \oplus MP_i^{new}$. Finally, the smart card replaces e_i by e_i^{new} in its memory.

B. The Protocol $\mathcal{P}_2$

In the proposed scheme $\mathcal{P}_1$, if an adversary $\mathcal{A}$ compromises a sensor node S_j and obtains the long-term secret f_j from it, $\mathcal{A}$ will be able to compute the previous session keys between U_i and S_j. Thus, we propose an improved version of $\mathcal{P}_1$ which can withstand the aforementioned weakness.

The advanced scheme $\mathcal{P}_2$ only differs from $\mathcal{P}_1$ in the authentication phase. Figure 7 illustrates the details of the authentication phase in $\mathcal{P}_2$. Here, we shows the modifications from $\mathcal{P}_1$ as follows:

- In Step 1 of the authentication phase, the smart card does not assign a random nonce to K_i, but it chooses a random number $a \in \mathbb{Z}_p$ and computes $K_i = aP$.
- In Step 4, the sensor chooses a random number $b \in \mathbb{Z}_p$ and computes $K_j = bP$.
- In the end, the user U_i and the sensor node S_j compute the session key $SK = h(abP.x)$, where $abP.x$ is the x-coordinate of the point abP.

We note that the differences in implementation between the two protocols $\mathcal{P}_1$ and $\mathcal{P}_2$ are insignificant. The changes in $\mathcal{P}_2$

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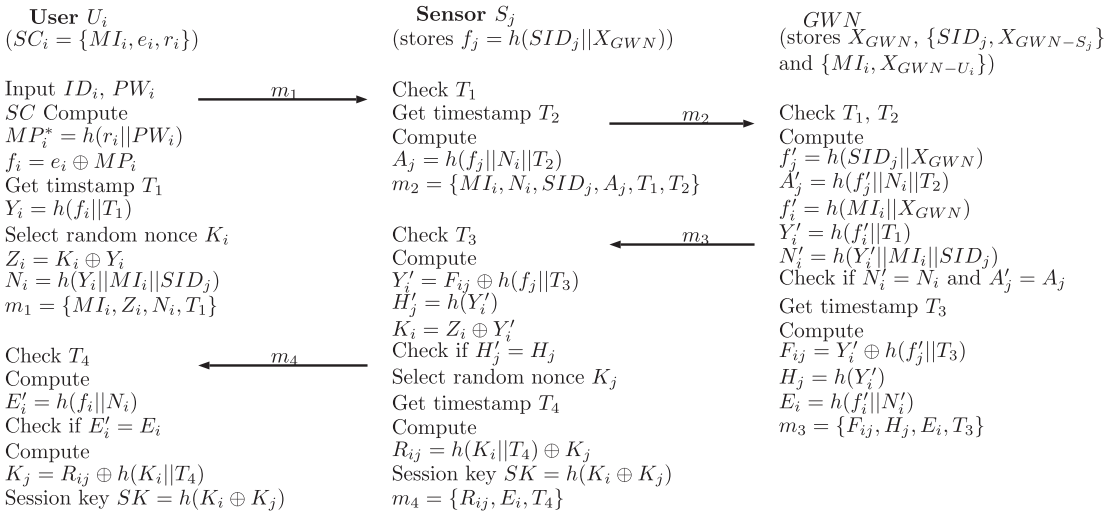


Fig. 6. Authentication phase in $\mathcal{P}_1$.

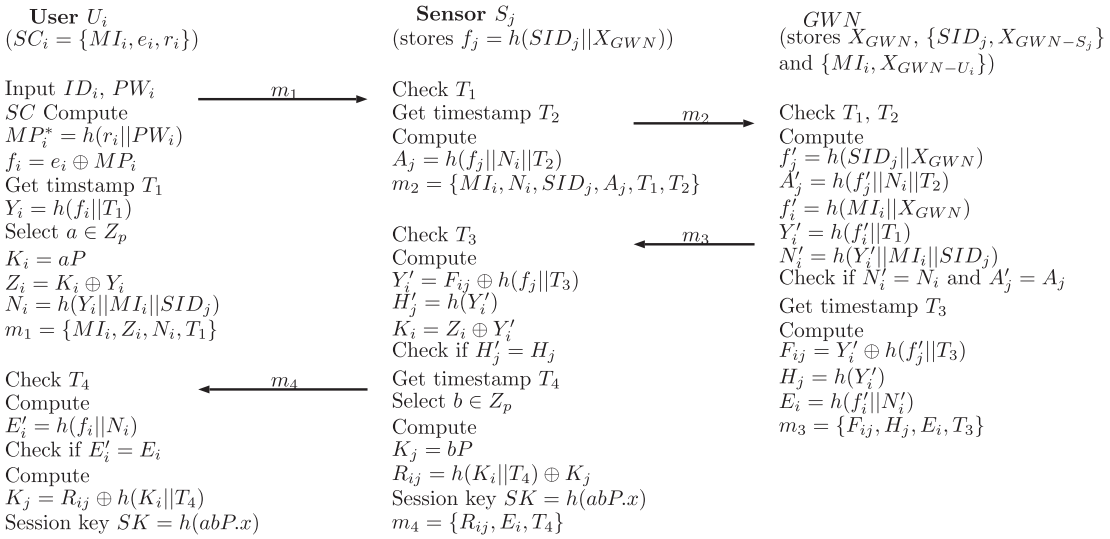


Fig. 7. Authentication phase in $\mathcal{P}_2$.

only affect the user and the sensor node, whereas the gateway node is left intact. When switching from one protocol to the other, the user and the sensor node just have to change the way they compute K_i, K_j and the session key SK.

V. Security Analysis

In this section, we show that the proposed protocols $\mathcal{P}_1$ and $\mathcal{P}_2$ are both secure. Moreover, the protocol $\mathcal{P}_2$ also provides perfect forward secrecy when the long-term secret of a sensor node is compromised.

Theorem 1: Let $\mathcal{A}$ be the adversary running in polynomial time t against our protocol $\mathcal{P}_1$ in random oracle model. Let D be a uniformly distributed password dictionary. Suppose that none of the sensor nodes have been compromised by the adversary. Then, we have:

$$Adv_{\mathcal{P}_1}^{ake}(\mathcal{A}) \leq \frac{q_h^2}{|Hash|} + \frac{2q_{send}}{|D|},$$

where q_h, q_{send}, $|Hash|$, and $|D|$ denote the number of $Hash$ queries, the number of $Send$ queries, the range space of the hash function, and the size of D, respectively.

Proof: In this proof, we define a sequence of games G_i, where $i = [0, 3]$. Let $Succ_i$ denote the event wherein the adversary succeeds in guessing the bit b in game G_i.

Game G_0: This game G_0 is the real attack by the adversary $\mathcal{A}$ against the protocol $\mathcal{P}_1$ in random oracle model. At the beginning of this game, the bit b is chosen at random. By definition, we have

$$Adv_{\mathcal{P}_1}^{ake}(\mathcal{A}) = 2Pr[Succ_0] - 1. \quad (1)$$

Game G_1: This game simulates of the adversary's eavesdropping attacks by querying $Execute(\Pi^t, \Pi^l)$ oracle. At the end, the adversary queries $Test$ oracle, it has to decide whether the output of $Test$ is the real session key SK or a random number. SK is computed from K_i and K_j. We know that $K_i = Z_i \oplus Y_i$, $Y_i = Z_i \oplus h(f_i \| T_1)$ or $Y_i = h(f_j \| T_3) \oplus F_{ij}$, where $f_i = h(MI_i \| X_{GWN})$ or $f_i = e_i \oplus h(r_i \| PW_i)$. Thus, the adversary

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cannot compute K_i from the transcripts unless it possesses both the smart card and password or corrupts the sensor node to obtain f_j.

Since $K_j = h(K_i \| T_4) \oplus R_{ij}$, the adversary must compute K_i before obtaining K_j. Thus, before corrupting the sensor node S_j and without both the smart card and the user's password, eavesdropping does not increase the chance of winning for the adversary in this experiment; so we have

$$Pr[Succ_0] = Pr[Succ_1]. \tag{2}$$

Game G_2: The differences between G_2 and G_1 is that we add the simulations of the *Send* and the *Hash* oracles. This game models an active attack in which the adversary $\mathcal{A}$ tries to deceive a participant into accepting a message fabricated by it. $\mathcal{A}$ queries the Hash oracle repeatedly to find collisions. Since each of the messages is associated to the identity of a participant and timestamp, there is no collision when querying the *Send* oracle. According to the birthday paradox, we have

$$|Pr[Succ_1] - Pr[Succ_2]| \leq q_h^2/2|Hash|. \tag{3}$$

Game G_3: This game G_3 simulates the $CorruptSC$ oracle and models the smart card lost attack. Since the password has low-entropy, the adversary might try online dictionary attack with the information obtained from the smart card; however, the number of wrong password inputs should be limited by the system. Thus, we have

$$|Pr[Succ_2] - Pr[Succ_3]| \leq q_{send}/|D|. \tag{4}$$

In the last game, all the oracles have been simulated. If the adversary has tried all the possible ways to break the security of the protocol $\mathcal{P}_1$ without any success at this point, then the last resource for winning the game after querying $Test()$ is guessing the bit b. Therefore, $Pr[Succ_3]$ equals to the probability of guessing the bit b, $Pr[Succ_3] = 1/2$.

From all the games, we have $Pr[Succ_0] = Pr[Succ_1]$, $|Pr[Succ_1] - Pr[Succ_2]| \leq q_h^2/2|Hash|$, $|Pr[Succ_2] - Pr[Succ_3]| \leq q_{send}/|D|$, and $Pr[Succ_0] = Adv^{ake}_{\mathcal{P}_1}(\mathcal{A})/2 + 1/2$. Thus,

$$Adv^{ake}_{\mathcal{P}_1}(\mathcal{A}) \leq \frac{q_h^2}{|Hash|} + \frac{2q_{send}}{|D|}.$$

■

Therefore, the proposed protocol $\mathcal{P}_1$ is secure when the range of the hash function and the size of the password dictionary are large. The adversary was allowed to query $CorruptSC$ oracle which simulates the stolen smart card and smart card breach attacks. We have shown that, even in the smart card lost scenario, the only way breaching the protocol is to mount online guessing attack. Since the authentication system normally limits the number of failed logins which is much smaller than $|D|$, the online password guessing attack is ineffective. Thus, the protocol $\mathcal{P}_1$ is immune to the stolen smart card and smart card breach attacks. Next, we show that the proposed protocol $\mathcal{P}_2$ is secure and it provides perfect forward secrecy as follows.

Theorem 2: Let $\mathcal{A}$ be the adversary running in polynomial time t against our protocol $\mathcal{P}_2$ in random oracle model. Let D be a uniformly distributed password dictionary. Then, we have:

$$Adv^{ake}_{\mathcal{P}_2}(\mathcal{A}) \leq \frac{q_h^2}{|Hash|} + \frac{2q_{send}}{|D|} + 2Adv^{ECDDH}_{G_p}(t),$$

where q_h, q_{send}, $|Hash|$, $|D|$, and $Adv^{ECDDH}_{G_p}(t)$ denote the number of $Hash$ queries, the number of *Send* queries, the range space of the hash function, the size of D, and the advantage of $\mathcal{A}$ in breaking the ECDDH problem, respectively.

Proof: In this proof, the games $G_0, G_1, \ldots, G_3$ are the same as in the proof for Theorem 1 except that $Pr[Succ_3] \neq 1/2$.

Game G_4: This game models the attack wherein the adversary has compromised the sensor node S_j by simulating the $CorruptSensor$ oracle. Thus, $\mathcal{A}$ can use the long-term secret f_j to compute K_i and K_j from a transcript. In the protocol $\mathcal{P}_2$, $K_i = aP$ and $K_j = bP$ and the session key $SK = h(abP.x)$. Let $Adv^{ECDDH}_{G_p}$ be the advantage of the adversary in the experiment wherein $\mathcal{A}$ has to distinguish between abP and a random number given aP and bP. Then, we have

$$|Pr[Succ_3] - Pr[Succ_4]| \leq Adv^{ECDDH}_{G_p}(t). \tag{5}$$

In the last game, it is clear that $Pr[Succ_4] = 1/2$.

Thus, from (1) to (5), we have $|Pr[Succ_0] - 1/2| \leq q_h^2/2|Hash| + q_{send}/|D| + Adv^{ECDDH}_{G_p}(t)$, and $Pr[Succ_0] = Adv^{ake}_{\mathcal{P}_2}(\mathcal{A})/2 + 1/2$. Hence,

$$Adv^{ake}_{\mathcal{P}_2}(\mathcal{A}) \leq \frac{q_h^2}{|Hash|} + \frac{2q_{send}}{|D|} + 2Adv^{ECDDH}_{G_p}(t).$$

■

The $CorruptSensor$ oracle in Game 4 simulates the attacks when a sensor node is captured by the adversary. In the case where sensor node compromising is highly probable, the protocol $\mathcal{P}_2$ is secure on the assumption that the Elliptic Curve Decisional Diffie-Hellman is computationally infeasible for a probabilistic polynomial time adversary as shown in Game 4. Therefore, the attacks related to sensor node capture are nullified in $\mathcal{P}_2$. The session key $SK = h(abP.x)$ is secure based on ECDDH even when leaking the ephemeral secrets of sensors or smart cards. Thus, $\mathcal{P}_2$ provides secure and perfect forward secrecy authentication.

VI. Performance Comparison

In this section, we evaluate the performances of our proposed protocols in terms of computation cost in the authentication phases. Table II shows the comparison between the protocol $\mathcal{P}_1$ and its lightweight protocols counterparts (Li et al. [11] and Turkanovic et al. [1]). Table III presents the comparison between the protocol $\mathcal{P}_2$, Shi et al. [15] and Choi et al. [16] which are the ECC based protocols.

Let t_{mul}, t_h and t_{XOR} be the time for performing an elliptic curve point multiplication, a hash operation, and an XOR operation, respectively. The time t_{mul} is about 0.063075 seconds,

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TABLE II
COMPARISON OF COMPUTATION COST WITH LIGHT WEIGHT PROTOCOLS

	Li et al. [11]	Turkanovic et al. [1]	$\mathcal{P}_1$
User	$9t_h + 6t_{XOR}$ $\simeq 0.0045s$	$7t_h + 4t_{XOR}$ $\simeq 0.0035s$	$7t_h + 4t_{XOR}$ $\simeq 0.0035s$
Sensor	$5t_h + 3t_{XOR}$ $\simeq 0.0025s$	$5t_h + 6t_{XOR}$ $\simeq 0.0025s$	$5t_h + 4t_{XOR}$ $\simeq 0.0025s$
Gateway	$12t_h + 6t_{XOR}$ $\simeq 0.006s$	$7t_h + 4t_{XOR}$ $\simeq 0.0035s$	$8t_h + 1t_{XOR}$ $\simeq 0.004s$

TABLE III
COMPARISON OF COMPUTATION COST WITH ECC-BASED PROTOCOLS

	Shi et al. [15]	Choi et al. [16]	$\mathcal{P}_2$
User	$3t_{mul} + 5t_h$ $\simeq 0.191725s$	$3t_{mul} + 9t_h$ $\simeq 0.193725s$	$2t_{mul} + 7t_h$ $\simeq 0.12765s$
Sensor	$2t_{mul} + 3t_h$ $\simeq 0.12765s$	$2t_{mul} + 6t_h$ $\simeq 0.12915s$	$2t_{mul} + 5t_h$ $\simeq 0.12865s$
Gateway	$1t_{mul} + 4t_h$ $\simeq 0.065075s$	$1t_{mul} + 5t_h$ $\simeq 0.065575s$	$9t_h$ $\simeq 0.0045s$

TABLE IV
COMPARISON OF SECURITY FEATURES

	[11]	[1]	$\mathcal{P}_1$	[15]	[16]	$\mathcal{P}_2$
Mutual authentication	Yes	Yes	Yes	Yes	Yes	Yes
Session key security	Yes	Yes	Yes	No	Yes	Yes
Perfect forward secrecy	No	No	No	Yes	Yes	Yes
Replay attack	Yes	No	No	No	No	No
Smart card lost attack	No	Yes	No	No	No	No
User anonymity	Yes	Yes	No	No	No	No
Impersonation attack	No	Yes	No	No	No	No
Stolen verifier attack	No	Yes	No	Yes	No	No

and t_h is about 0.0005 seconds [14]. t_{XOR} is very small compared to t_{mul} and t_h so we neglect it when computing the time approximately.

Among the lightweight protocols, Li et al.'s has worst computation cost. Table II shows that our protocol $\mathcal{P}_1$ is slightly more efficient than Turkanovic et al.'s scheme at sensor node.

For the ECC-based protocols in Table III, our protocol $\mathcal{P}_2$ outperforms Shi et al.'s and Choi et al.'s schemes. At the user side, $\mathcal{P}_2$ executes 1 ECC point multiplication less than the other's schemes in each user authentication cycle. While the gateway node in Shi et al.'s and Choi et al.'s protocols computes 1 point multiplication, it only computes hash operations and runs approximately 14 times faster in $\mathcal{P}_2$.

Table IV presents the security feature comparison. The Choi et al.'s and $\mathcal{P}_2$ are the most secure schemes, but Table III shows that $\mathcal{P}_2$ is more efficient than Choi et al's. Although $\mathcal{P}_1$ does not provide forward secrecy, it provides more security features than Li et al.'s, Turkanovic et al.'s, and Shi et al.'s schemes.

VII. CONCLUSION

In this paper, we review and show that Turkanovic et al.'s scheme is vulnerable to various kinds of attacks. First, an adversary can log into a sensor node with a stolen smart card. Second, the adversary can impersonate a user to access to any sensor node or mount sensor node spoofing attack after compromising a sensor node. Moreover, the adversary can use the aforementioned attacked to obtain the past and the future session keys. Lastly, Turkanovic et al.'s is also subject to stolen verifier attack, which allows the adversary to masquerade as a sensor node.

In order to counter those problems, we propose two authentication schemes for the wireless sensor networks. The first scheme overcomes those weaknesses and it is as efficient as the Turkanovic et al.'s. The second scheme is a slightly modified version of the first and it can provide perfect forward secrecy. The latter is suitable for being used in the extreme hostile environment where node capture occurs with high probability. Since the modification is insignificant, both schemes can be implemented on a sensor node with little storage cost. One can easily switch between the two protocols depending on the security requirements.

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